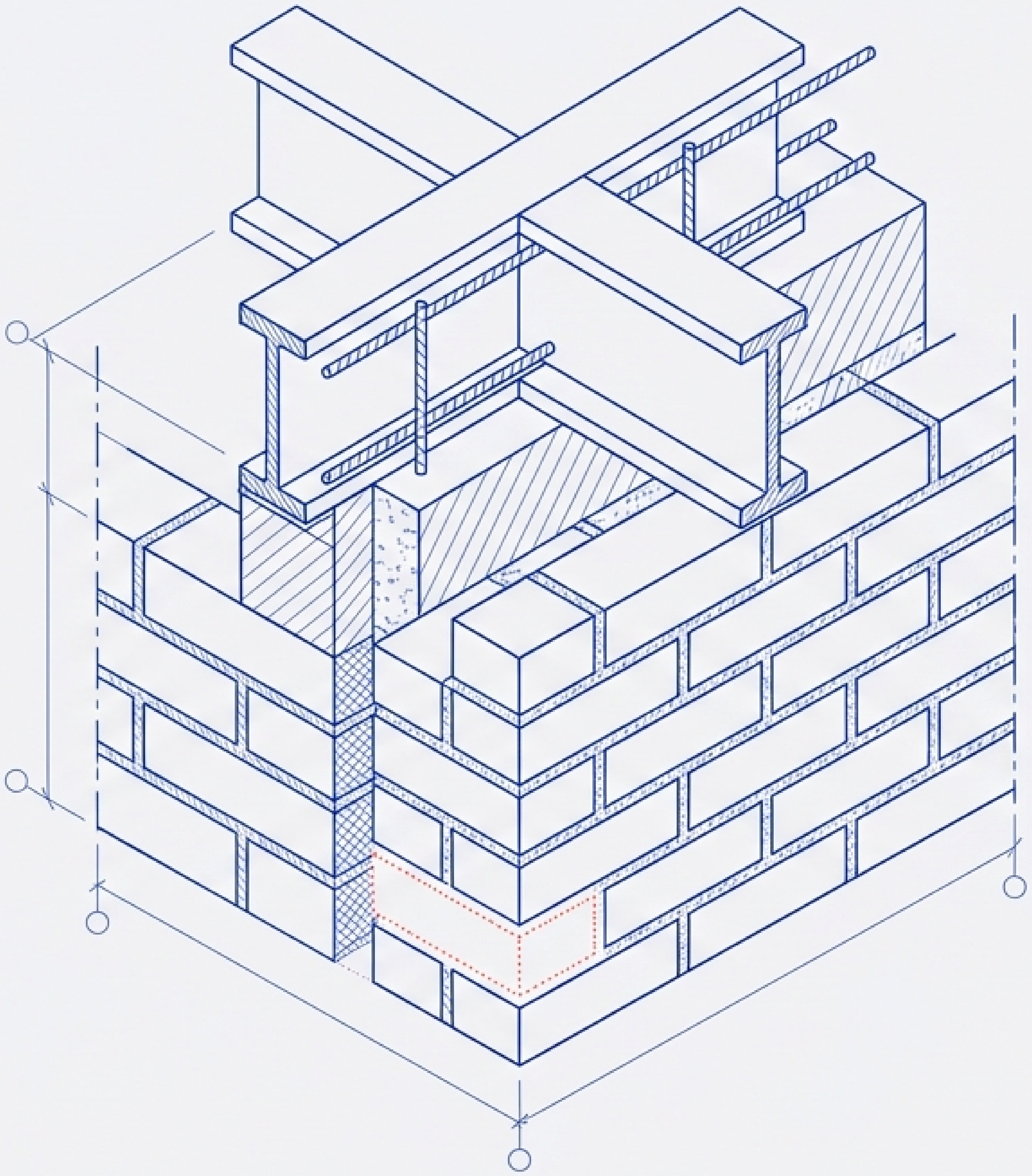
The Strategic Protocol: Data Hygiene & Complete Case Analysis

A standard for assessing, removing, and validating missing data in machine learning pipelines.



Handling Missing Data Series | Part 1: Removal Techniques

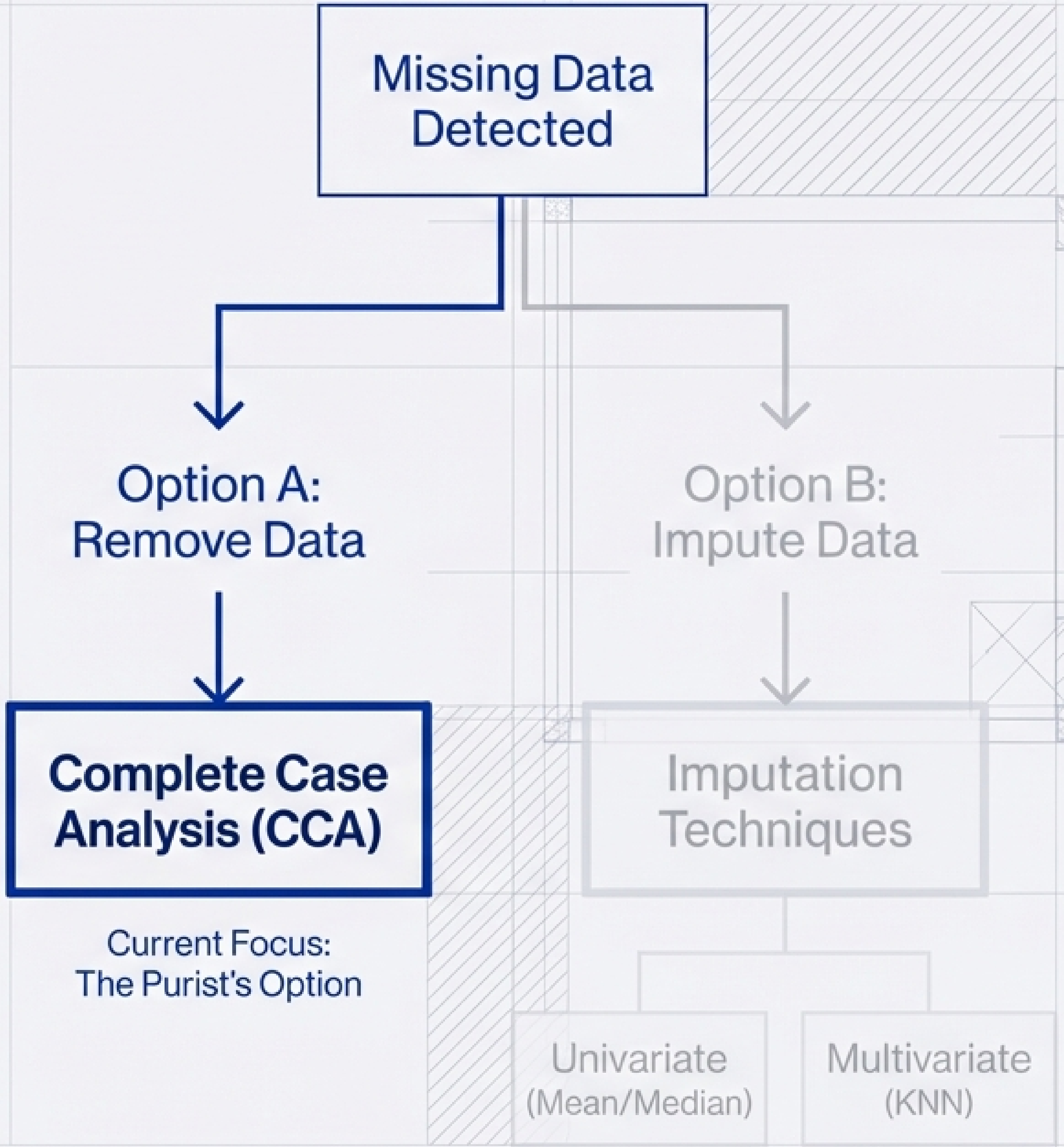
The Data Scientist's Binary Dilemma

The Constraint

Most Machine Learning algorithms cannot process missing data. Passing "NaN" values to a model will cause training failure.

The Responsibility

It is the Data Scientist's duty to clean the dataset before the model sees it.



Defining the Protocol: Complete Case Analysis (CCA)

Also known as Listwise Deletion.

	Feature A	Feature B	Feature C	Target
Case 1	1.2	3.5	8.9	0
Case 2	4.1	2.2	7.6	1
~~Case 3~~	~~5.5~~	Ø	~~9.1~~	~~1~~
Case 4	3.3	4.8	6.7	0
Case 5	2.9	3.0	5.2	1

Single Missing Value
= Row Deletion

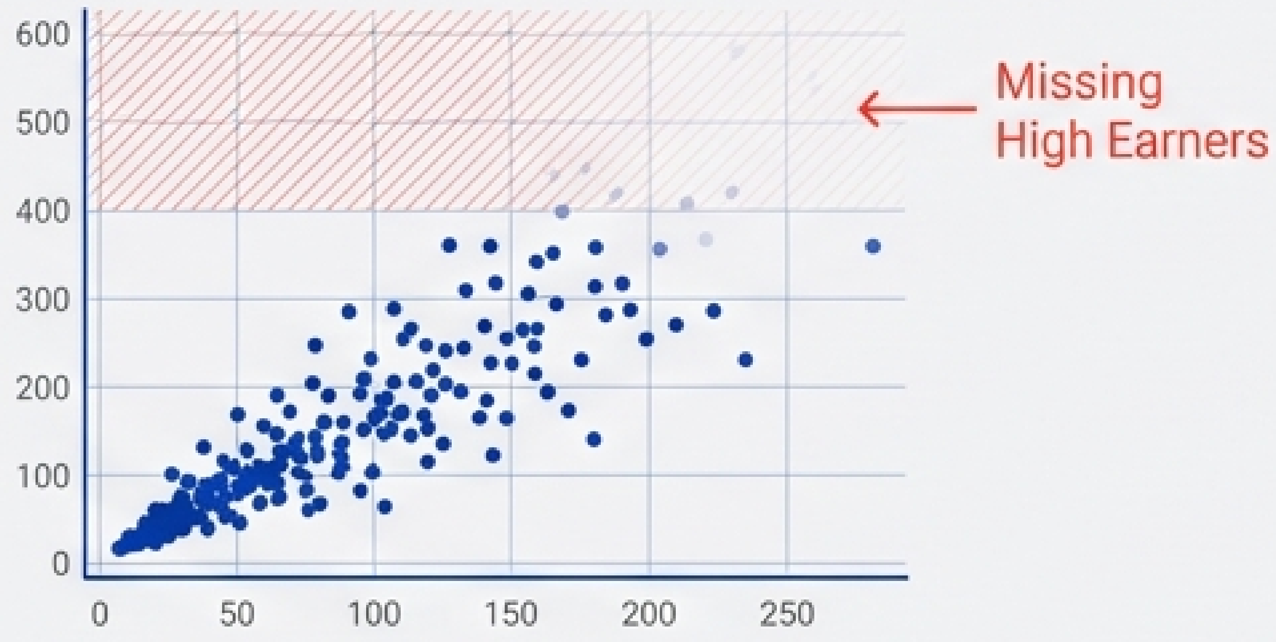
The Logic: The algorithm only analyses 'Complete Cases'—rows where information exists for every single column. If a value is missing, the row dies. It is a brutal but effective filter.

The Critical Assumption: Missing Completely At Random (MCAR)

CCA is only valid if the data is missing randomly. There must be no pattern to the missing values.

Non-Random Missingness (Bias)

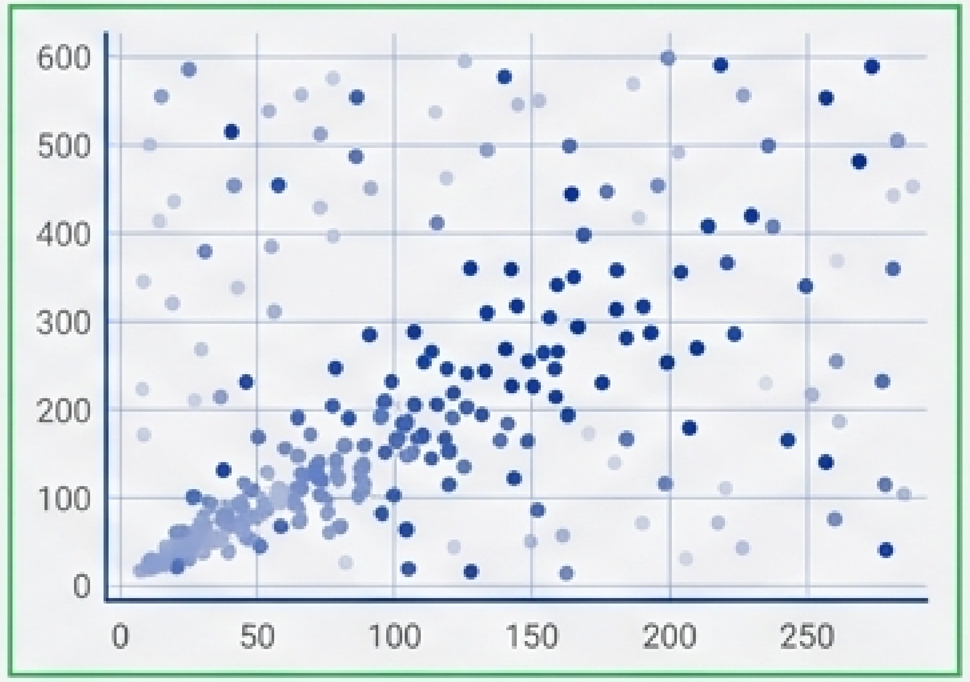
Scenario: High-income individuals intentionally hide their salary.



Result: The model fails to learn about high earners. Deletion destroys validity.

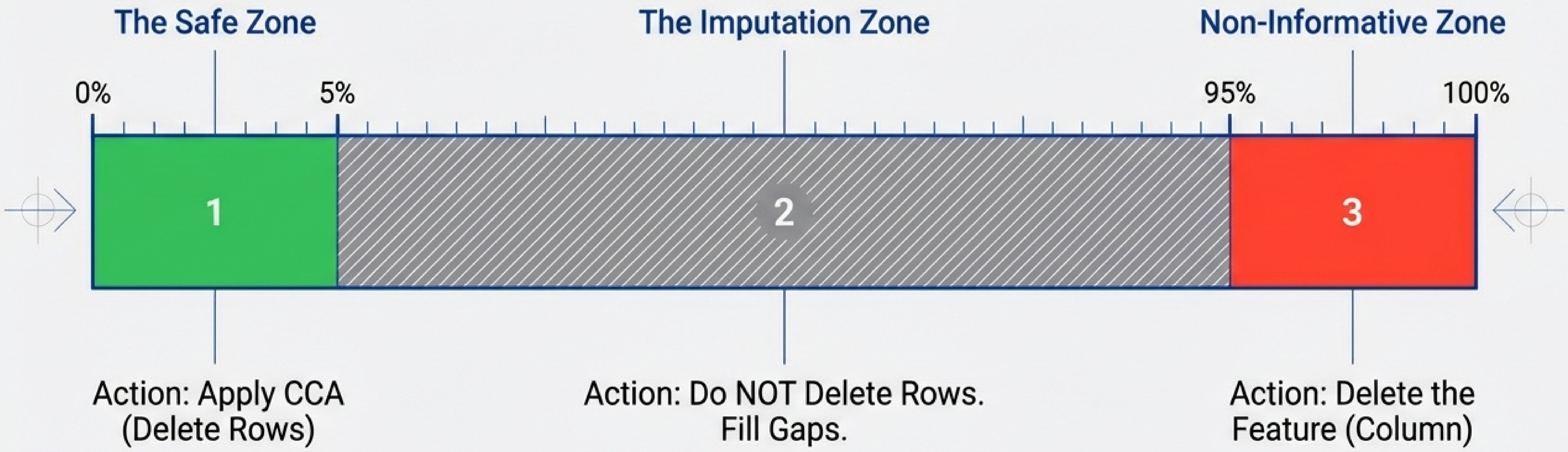
Missing Completely At Random (MCAR)

Scenario: Data lost due to a random server glitch affecting arbitrary users.



Result: Removal acts as a random sample. Distribution remains true.

Operating Thresholds: The 5% Heuristic



Standard Rule: CCA is generally considered safe if the missing data constitutes < 5% of the total dataset for that feature. If a column is missing >95%, the feature itself is non-informative.

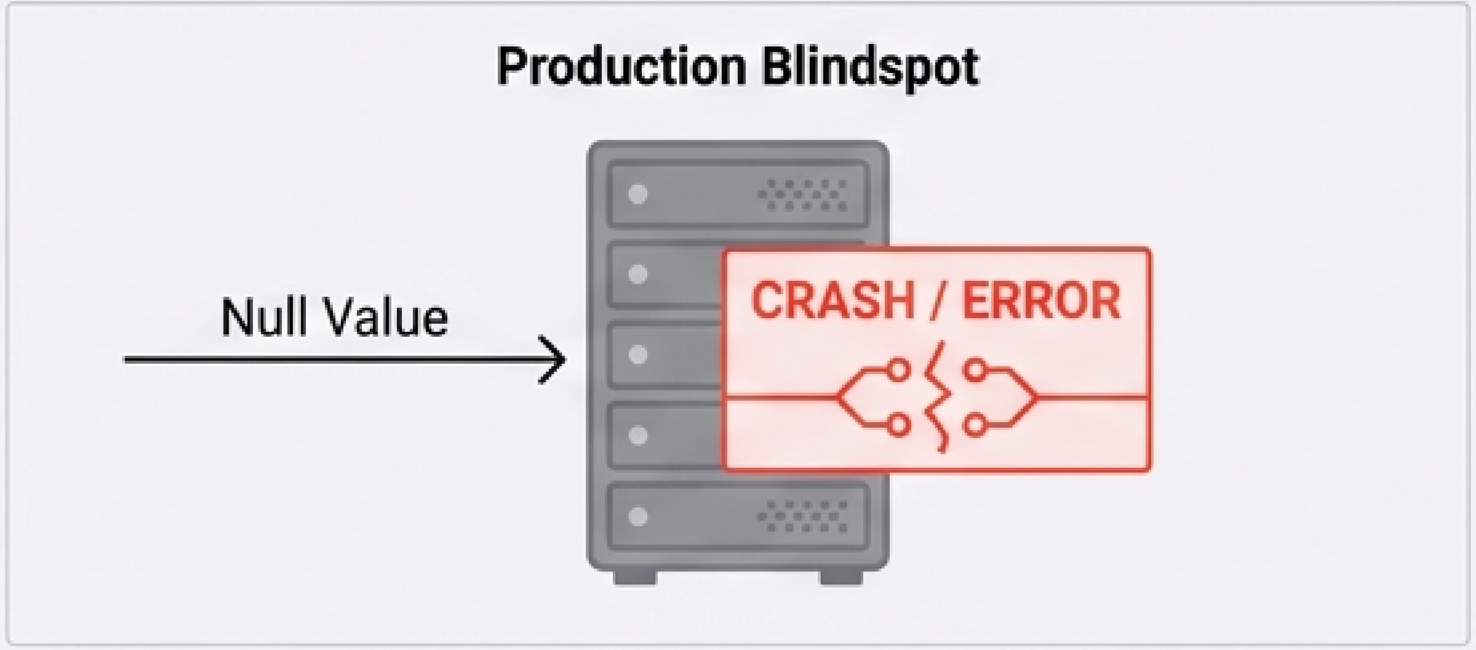
The Trade-off Assessment

Advantages ✅

1. **Simplicity**: No complex data manipulation or expensive computation required.
2. **Fidelity**: Preserves the exact distribution of the data (provided the MCAR assumption holds).

Risks 🚩

1. **Data Loss**: Potentially discarding useful information from other columns.
2. **The Production Blindspot** ⚠️



Warning: If trained only on complete cases, the model has no logic to handle missing data in a live production environment.

Case Study: The ‘Art & Science Jobs’ Dataset

Predicting candidate hiring (Target: 0 or 1)

Data Manifest

Numerical Variables

- Enrollee ID
- City Development Index
- Experience
- Training Hours

Categorical Variables

- City
- Gender
- Relevant Experience
- Enrolled University
- Education Level
- Major Discipline
- Company Size
- Company Type

Objective: Perform a Complete Case Audit to determine eligibility for row deletion.

The Audit: Identifying Candidates for CCA

Filtering features based on < 5% missingness threshold.

REJECTED (Do Not Delete Rows)

- Company_Type: ~32% Missing
- Company_Size: ~30% Missing
- Gender: ~23% Missing
- Major_Discipline: ~15% Missing

Too much data loss. Requires Imputation.

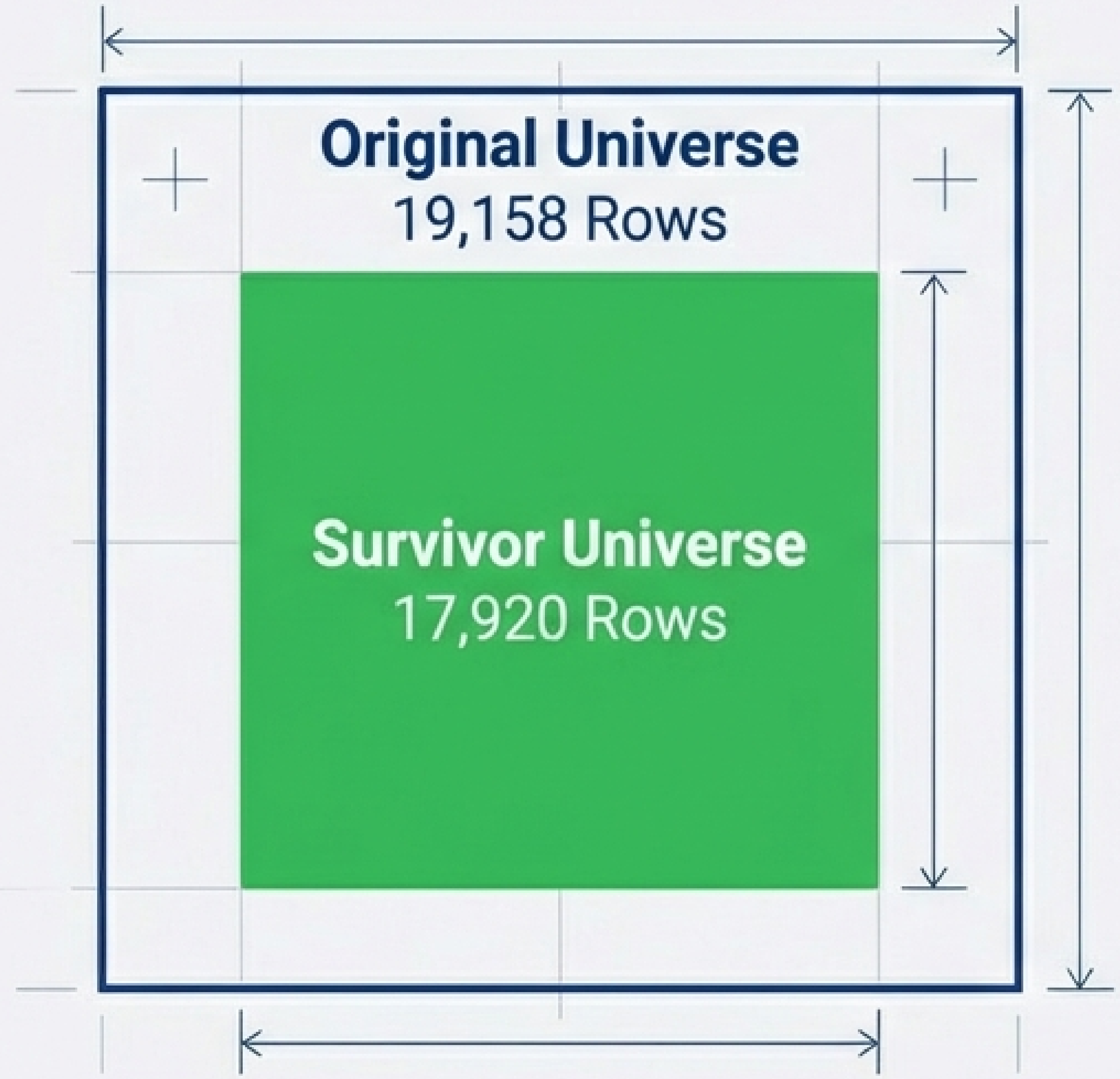
CANDIDATES (Safe for CCA)

- Training_Hours: ~4% Missing
- City_Development_Index: ~2% Missing
- Enrolled_University: ~2% Missing
- Education_Level: ~2% Missing
- Experience: ~0.3% Missing

Proceed with Row Deletion.

Execution: The Impact of Deletion

Methodology: Applied deletion on the 5 candidate columns.

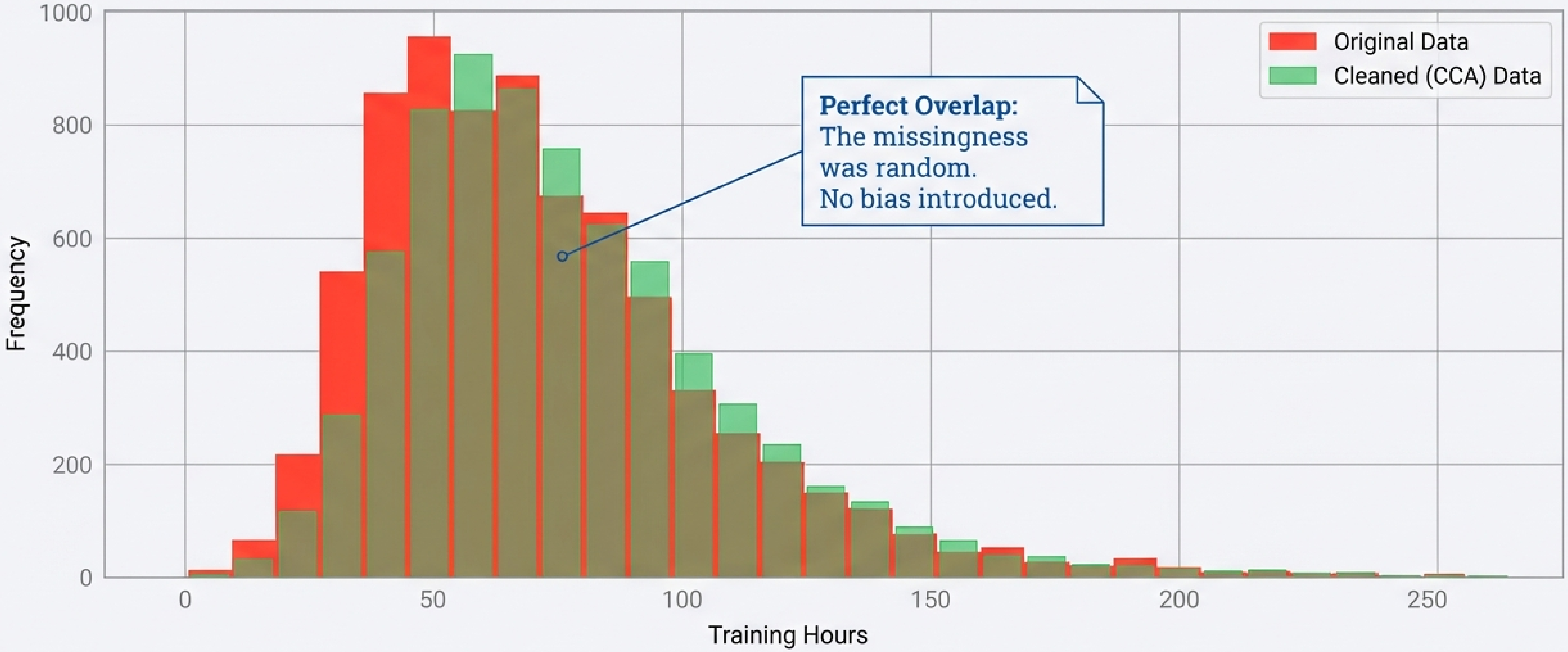


93.5% Retention Rate

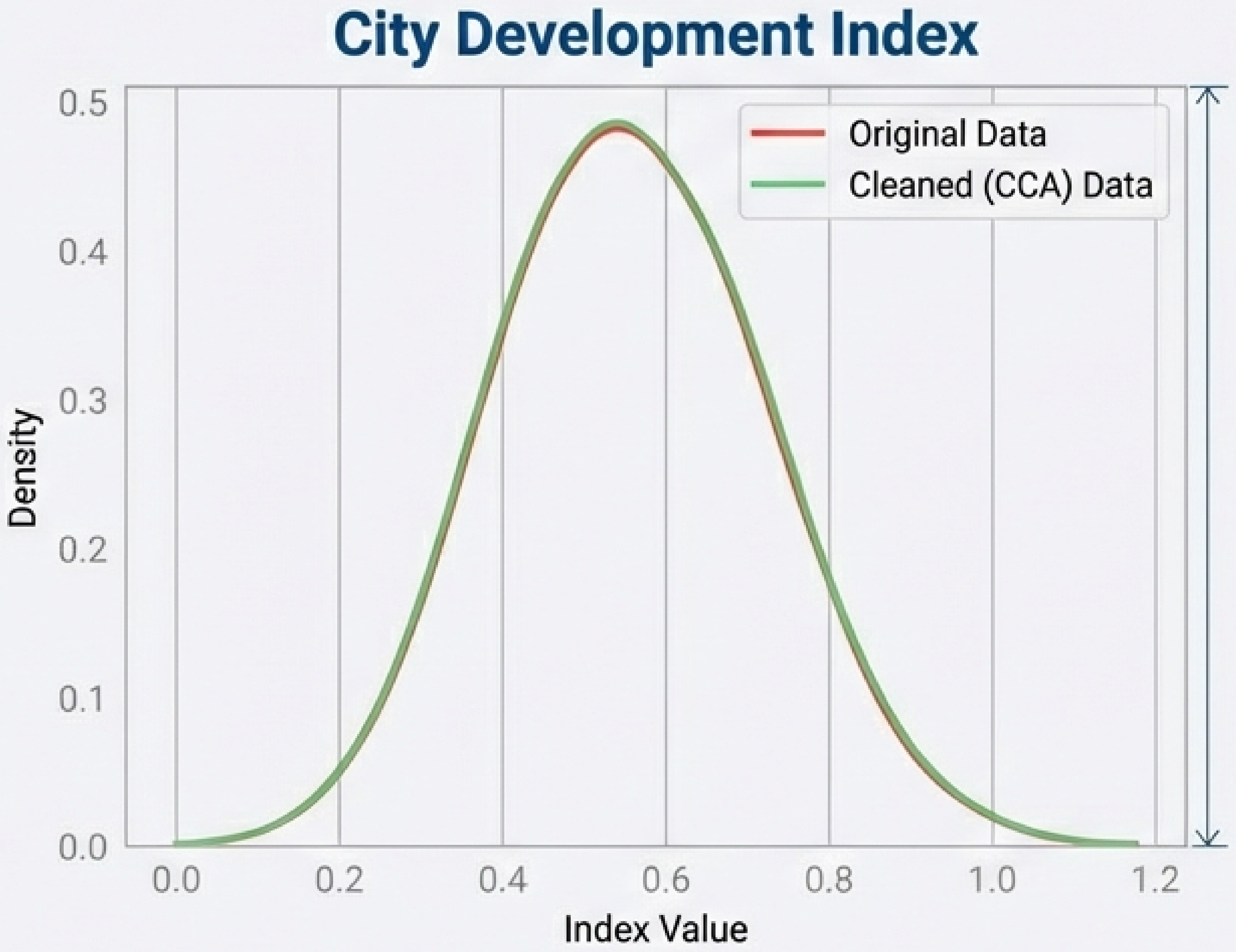
This falls well within acceptable limits. High-missingness columns (e.g., Gender) were excluded from the drop operation to preserve volume.

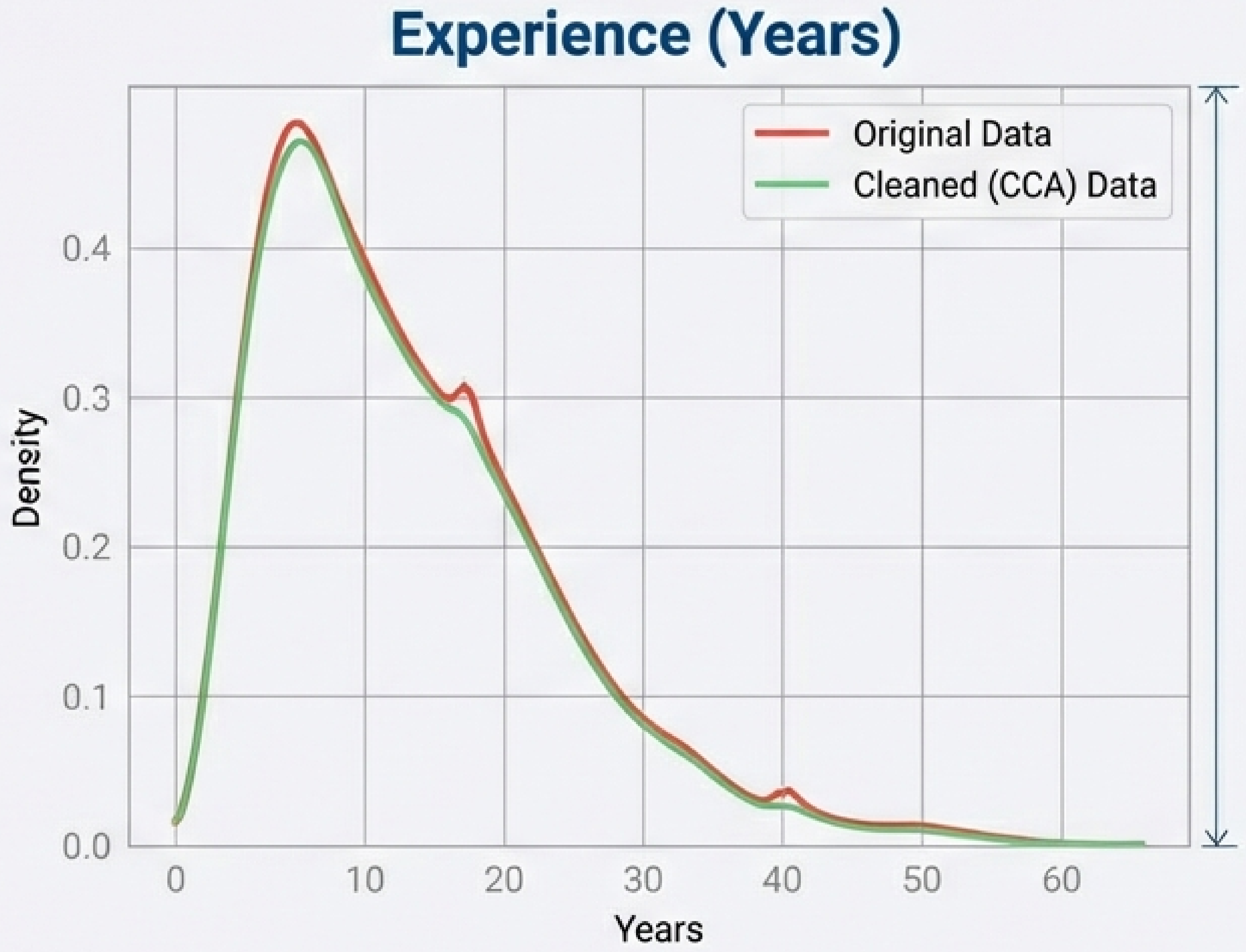
Validation (Numerical): Training Hours Distribution

Overlaying histograms to check for distribution shift.



Validation (Numerical): Probability Density Function (PDF)

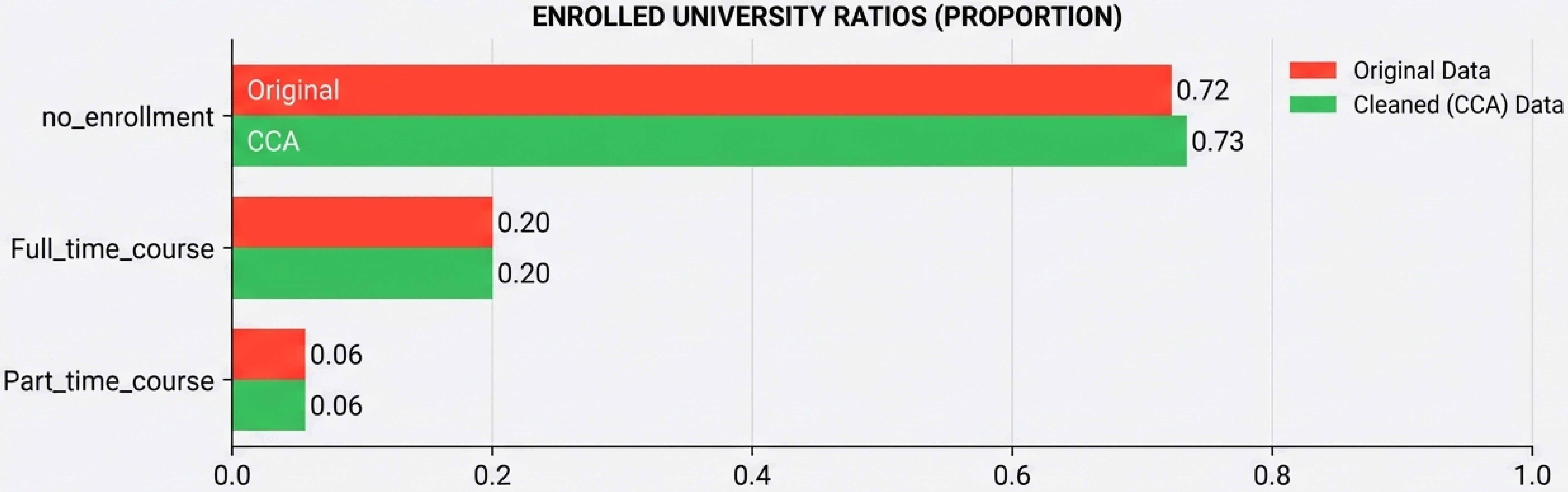




Verdict: The data structure is intact. Numerical audit passed.

Validation (Categorical): Enrolled University Ratios

Checking category proportions pre- and post-deletion.



Ratios are virtually unchanged. If "no_enrollment" had shifted significantly (e.g., to 0.85), it would indicate systematic bias. That did not happen here.

Validation (Categorical): Education Level Stability

EDUCATION LEVEL	ORIGINAL		CCA
Graduate	0.60	=	0.61
Masters	0.22	=	0.23
High School	0.10	=	0.10
PhD / Primary		Static	

Proportions survived the deletion process. This confirms that missing data in "Education_Level" was MCAR (e.g., PhDs were not shier about sharing their status than Graduates).

The Verdict: Checklist for CCA Success

Volume Check
Is the missing data < 5% of the total?

Randomness Check (MCAR)
Is the data missing completely at random? No hidden patterns?

Distribution Check
Do histograms and category ratios match pre- and post-deletion?

Status: PASSED.

CCA is the **efficient, correct choice** for this subset.

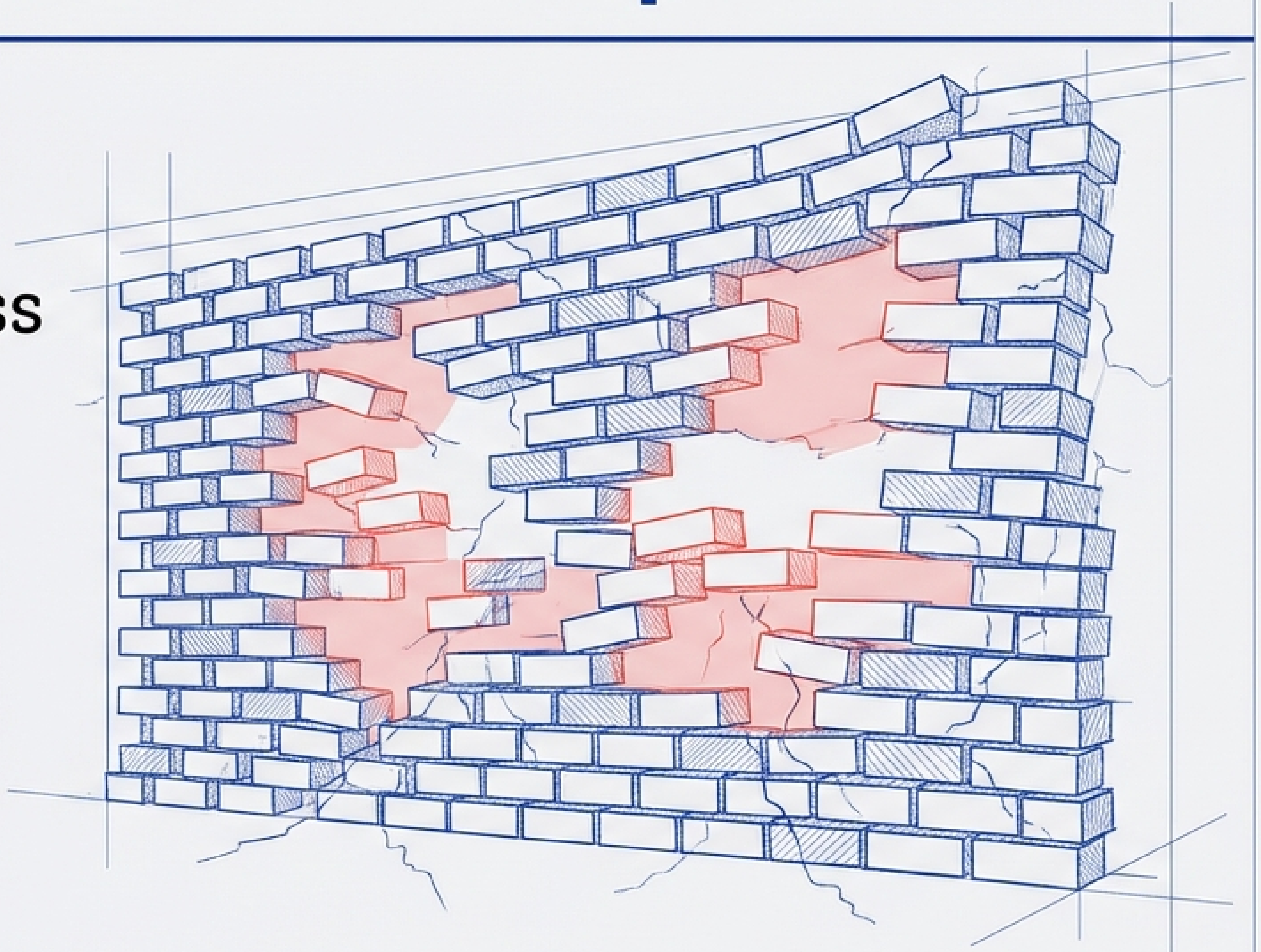
When Deletion Is Not An Option

The Reality:

In many datasets, missingness exceeds 5% or follows a pattern (Not MCAR).

The Consequence:

In these cases, deletion is destruction. We cannot remove the rows.



Next in the Protocol: Univariate & Multivariate Imputation Techniques.